

B.Tech. Degree VII Semester Special Supplementary Examination March 2023

ME 19-205-0703 MACHINE DESIGN II (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the customers need, formulate the problem and draw the design specifications.
 CO2: Understand component behavior subjected to loads and identify the failure criteria.
 CO3: Apply the concepts of stress analysis, theories of failure and material science to analyze, design and/or select commonly used machine components.
 CO4: Analyze the stresses and strains induced in a machine element.
 CO5: Design machine components using theories of failure.
 CO6: Get basic knowledge of the preparation of working drawings by including the manufacturing details like tolerance, surface finish etc.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

(Use of Design Data Handbook is permitted)

(Data not given may be suitably assumed. All such assumptions must be clearly stated)

PART A

(Answer **ALL** questions)

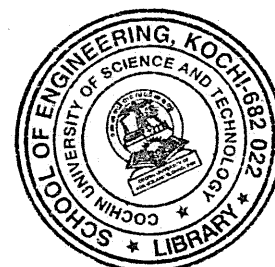
	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) What are the materials used for lining of friction surfaces?	3	L1	2	1	
(b) What is a self-energizing brake? When a brake becomes self-locking?	3	L1	2	1	
(c) Explain the phenomenon of interference in involute gears. What are the conditions to be satisfied in order to avoid interference?	3	L2	1	1,2	
(d) For bevel gears, define the following:	3	L1	2	1,2	
(i) Cone distance					
(ii) Pitch angle					
(iii) Face angle					
(iv) Root angle.					
(e) Explain the following terms as applied to journal bearings:	3	L1	2	1	
(i) Bearing characteristic number					
(ii) Bearing modulus.					
(f) What are rolling contact bearings? Discuss their advantages over sliding contact bearings.	3	L2	1	1	
(g) Describe about tolerance and surface finish.	3	L1	6	1,2	
(h) What are the points to be considered while designing a part which needs machining?	3	L2	6	1,2	

PART B

(4 × 12 = 48)

- II. A multiple disc clutch, steel on bronze, is to transmit 4.5 kW at 750 r.p.m. The inner radius of the contact is 40 mm and outer radius of the contact is 70 mm. The clutch operates in oil with an expected coefficient of 0.1. The average allowable pressure is 0.35 N/mm². Find:
- the total number of steel and bronze discs
 - the actual axial force required
 - the actual average pressure
 - the actual maximum pressure.

OR



(P.T.O.)

		Marks	BL	CO	PO
III.	Two pulleys, one 450 mm diameter and the other 200 mm diameter, on parallel shafts 1.95 m apart are connected by a crossed belt. Find the length of the belt required and the angle of contact between the belt and each pulley. What power can be transmitted by the belt when the larger pulley rotates at 200 rev/min, if the maximum permissible tension in the belt is 1 kN, and the coefficient of friction between the belt and pulley is 0.25?	12	L4	4	1,2,3
IV.	A gear drive is required to transmit a maximum power of 22.5 kW. The velocity ratio is 1:2 and speed of the pinion is 200 r.p.m. The approximate centre distance between the shafts may be taken as 600 mm. The teeth has 20° stub involute profiles. The static stress for the gear material (which is cast iron) may be taken as 60 MPa and face width as 10 times the module. Find the module, face width and number of teeth on each gear. Check the design for dynamic and wear loads. The deformation or dynamic factor in the Buckingham equation may be taken as 80 and the material combination factor for the wear as 1.4.	12	L4	5	1,2,3
OR					
V.	Design 20° involute worm and gear to transmit 10 kW with worm rotating at 1400 r.p.m. and to obtain a speed reduction of 12:1. The distance between the shafts is 225 mm.	12	L4	3	1,2,3
VI.	Design a main bearing for the compressor to sustain a load of 4.5 kN. The speed is 1500 rpm. Ambient temperature = 25°C. Diameter of the main shaft = 50 mm.	12	L3	3	1,2,3
OR					
VII.	A bearing for an axial compressor is to carry an axial thrust of 1500 N and a radial load of 2500 N. The bearing rotates at 1000 rpm and has to work 40 hours/week for 5 years. Select a suitable bearing for a shaft diameter of 50 mm.	12	L5	5	1,2,3
VIII.	(a) What are the general design recommendations for welded parts? Explain with sketches.	8	L2	6	1,2,3
	(b) What are the main design considerations for the parts produced in milling machines?	4	L2	6	1
OR					
IX.	(a) What are the important points to be considered while designing parts to be manufactured by forging? Explain with sketches.	8	L2	6	1,2,3
	(b) What are the general design recommendations for rolled sections?	4	L2	6	1

Blooms's Taxonomy Levels

L1 – 12.50%, L2 – 27.50%, L3 – 20%, L4 – 30%, L5 – 10%.
